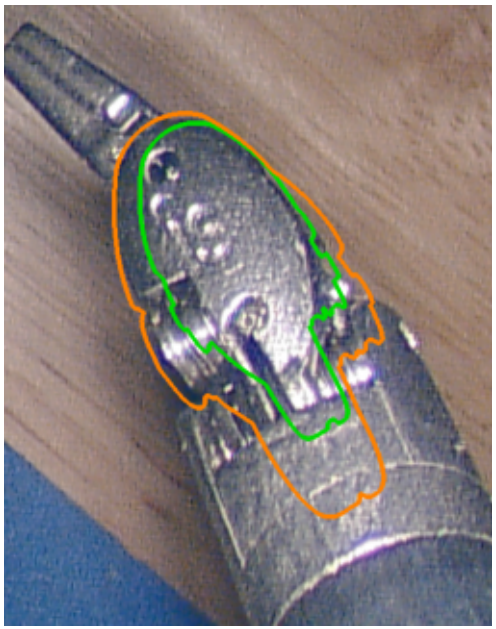
gt_dataset: diff-render (markerless) as a 6D pose estimator -- last training frame 14 (used in the fit) scored against the ground-truth pose (marker + 3D pose alignment)

diff-render (markerless): translation error 17.06 mm   rotation error 6.30 deg   ADD 17.04 mm   IoU 0.584



zoom: outlines GT (green) vs diff-render (orange)



SurgRIPE metrics (evaluate.py), frame 14

Metric	diff-render
ADD (mm)	17.04
ADD < 10% diam. (1.62 mm)	fail
ADD-S (mm)	12.88
ADD-S < 10% diam.	fail
2D proj. error (px)	37.9
2D proj. < 5 px	fail
5 mm / 5 deg	fail
translation error (mm)	17.06
rotation error (deg)	6.30
Avg Acc (0-5 mm)	0.000
ADD AUC (0-diam.)	0.000
silhouette IoU vs GT *	0.584

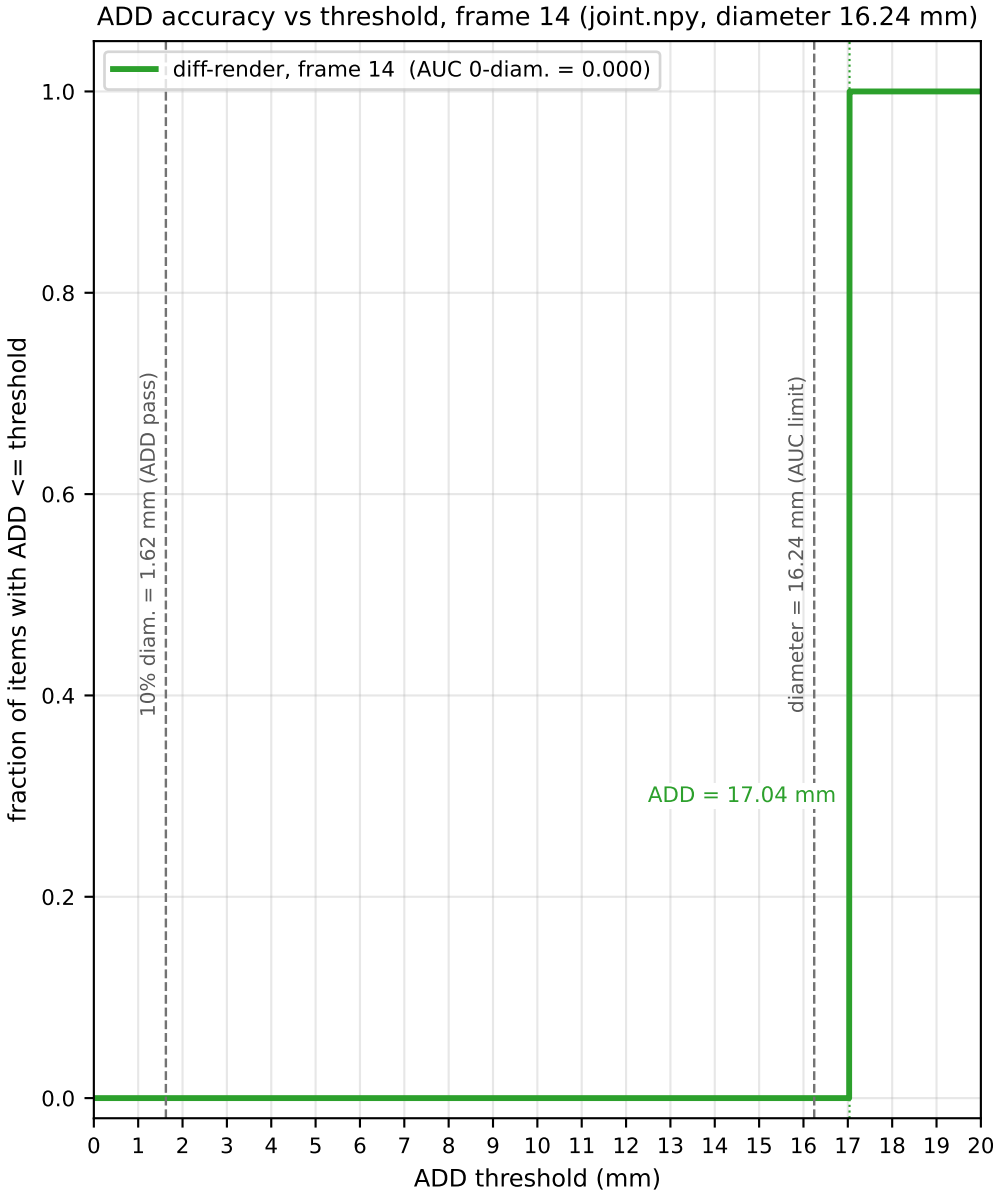
Where the error lies (camera frame), frame 14

	value
clevis depth: GT / diff-render (mm)	76.2 / 59.4
translation along line of sight (mm)	-16.95
translation sideways (mm)	1.88
rotation, total (deg)	6.30
about clevis long axis (roll)	6.25
about line of sight	1.63
projected clevis size: GT / DR (px)	317 / 405
apparent scale DR vs GT	1.28x

Negative along-sight = diff-render places the clevis closer to the camera than the ground truth. A depth error mostly changes the silhouette's SIZE, which the filled full-frame overlay above hides; the zoomed outlines show it. Roll about the clevis's long axis barely changes its near-symmetric silhouette.

Frame 14 is the last image the markerless diff-render calibration was fit on; the calibration is applied to its kinematics and compared with the ground truth applied the same way (ground truth used only for scoring). Metrics: SurgRIPE's evaluate.py Evaluator (ADD / ADD-S pass at 10% of the joint.npy diameter, 2D projection < 5 px with this frame's rectified K instead of SurgRIPE's fixed K, 5 mm 5 deg, Avg Acc 0-5 mm as in the SurgRIPE paper's Table 4).
* IoU is not a SurgRIPE metric. One image -> pass rates are 0/1.

gt_dataset: diff-render (markerless) as a 6D pose estimator -- last training frame 14 (used in the fit)
scored against the ground-truth pose (marker + 3D pose alignment)



A single image is one item, so the curve is a step at that image's ADD: 0 below it, 1 at and above it. SurgRIPE's AUC integrates the curve from 0 to the object diameter, so it is 0 whenever ADD exceeds the diameter.

Calibration run time (s): 15 training images, RTX 4060

frame	load + stereo box	masks	joint DR share	total / image
0	1.50	10.52	0.84	12.85
1	1.61	12.58	0.84	15.03
2	1.58	8.81	0.84	11.22
3	1.93	5.80	0.84	8.57
4	1.40	14.10	0.84	16.34
5	1.45	7.80	0.84	10.08
6	1.65	7.37	0.84	9.86
7	1.56	9.05	0.84	11.44
8	1.52	13.26	0.84	15.62
9	1.46	13.42	0.84	15.72
10	1.41	7.64	0.84	9.89
11	1.25	6.16	0.84	8.24
12	1.23	7.57	0.84	9.64
13	1.34	6.59	0.84	8.77
14	1.52	8.91	0.84	11.26
mean / image	**1.49**	**9.31**	**0.84**	**11.64**
TOTAL (timed)				**180.2**
TOTAL (this fit)				**150.9**

Joint steps (all 15 fit images at once, split equally above): edge LUT 0.0, PnP init 0.11, DR passes 3.1 + 2.7 + 2.6 + 2.7, outlier check 1.4, refit 0.0 = 12.5 s. Python overhead 5.6 s is in TOTAL only. One-time start-up (imports, CUDA, mesh, renderer) 1.8 s extra -> 182.0 s from launch. The fit in this report predates run-time logging (only its total was recorded); the per-image times are from a separate timed fit of the same 15 images (2026-11-03T18:12:33).

gt_dataset: hand-eye (marker only) vs diff-render (markerless), both scored against the ground truth (marker + 3D pose alignment)

Held-out frames [15, 16, 17, 18, 19] (fit on frames [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14]; held-out images never seen by the fit)

Frame	hand-eye trans. err (mm)	hand-eye rot. err (deg)	hand-eye IoU	diff-render trans. err (mm)	diff-render rot. err (deg)	diff-render IoU
15	15.21	1.10	0.103	16.88	6.30	0.589
16	15.22	1.10	0.151	16.84	6.30	0.572
17	15.22	1.10	0.101	16.82	6.30	0.531
18	15.20	1.10	0.072	16.91	6.30	0.579
19	15.19	1.10	0.082	17.20	6.30	0.558
Mean	**15.21**	**1.10**	**0.102**	**16.93**	**6.30**	**0.566**

Whole calibration, averaged over the held-out frames

Calibration	mean clevis displacement (mm)	rotation err (deg)	base-origin err (mm)
PnP init (no prior)	41.66	34.79	33.24
hand-eye (marker only)	15.17	1.10	15.82
diff-render (markerless)	16.91	6.30	17.33

trans. err = distance between the clevis origin (pitch pin) under the calibration and under the ground truth; rot. err = geodesic angle between the two clevis orientations (equal to the calibration's rotation error, so the same in every frame); IoU = overlap of the clevis silhouettes rendered at the calibration and at the ground truth. Each calibration is applied to the held-out frame's kinematics; nothing is fit per frame. Diff-render inputs (training frames only): rectified images, camera/stereo calibration, joint.stl, PSM kinematics -- no ground truth, marker calibration or initial pose.

```
================================================================================================================
    hand_eye_diff_render_comparison.py — markerless hand-eye calibration by differentiable rendering
================================================================================================================
Dataset   : /home/aseshad4/hand_eye_collection_scripts/gt_dataset  (20 frames)
Fit on    : frames [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14] (first 75%; tip found in 15, 15 kept after outlier rejection)
Held out  : frames [15, 16, 17, 18, 19] -- never opened by the fit; the only frames scored
Generated : 2026-11-04T16:21:08   fit runtime 151s
Ground truth = marker registration + 3D pose alignment (cam_T_base_post_alignment); used ONLY for scoring.
DR inputs   = rectified right+left images, intrinsics / rect_T_cam_raw / stereo P, joint.stl, PSM kinematics.
              No marker calibration, no ground truth, no initial pose, no annotation.
Pipeline    = stereo tool box -> GrabCut -> PnP init -> 4 DR passes (masks re-seeded from the
              DR estimate after pass 1) -> outlier rejection + refit.

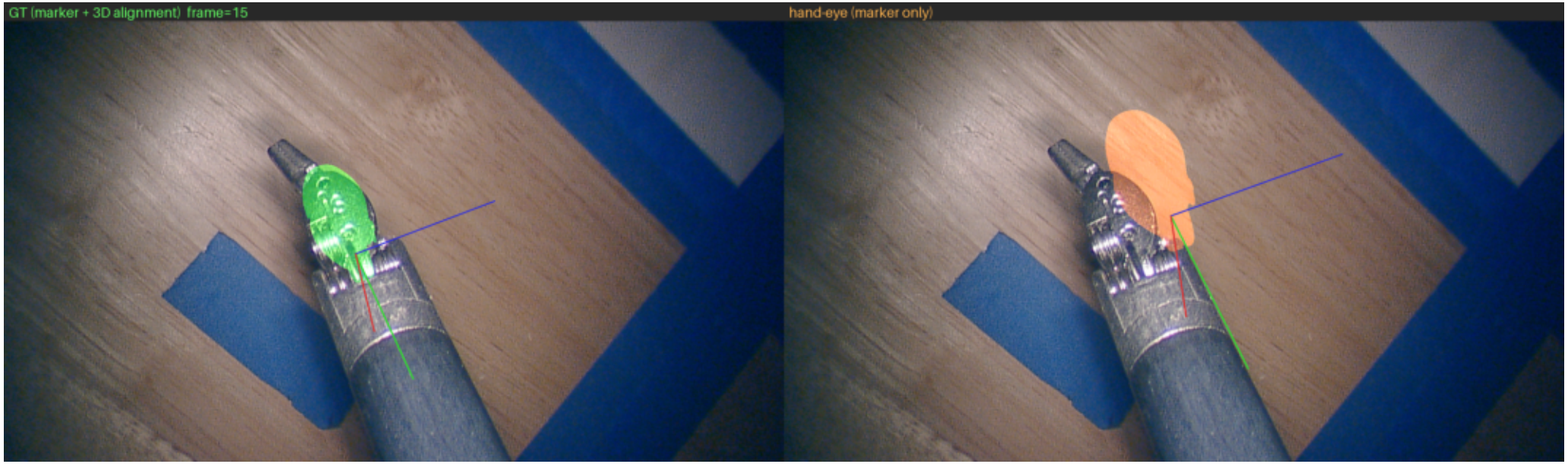
Whole calibration vs ground truth (held-out frames):
                              clevis displ. (mm)  rotation (deg)  base origin (mm)
  PnP init (no prior)                      41.66           34.79             33.24
  hand-eye (marker only)                   15.17            1.10             15.82
  diff-render (markerless)                 16.91            6.30             17.33

Held-out frames (calibration applied to each frame's kinematics vs the ground truth applied the same way; IoU of the rendered silhouettes):
  frame    hand-eye (mark t(mm)  rot(deg)    IoU   diff-render (m t(mm)  rot(deg)    IoU
  ------------------------------------------------------------------------------------
     15                   15.21      1.10  0.103                 16.88      6.30  0.589
     16                   15.22      1.10  0.151                 16.84      6.30  0.572
     17                   15.22      1.10  0.101                 16.82      6.30  0.531
     18                   15.20      1.10  0.072                 16.91      6.30  0.579
     19                   15.19      1.10  0.082                 17.20      6.30  0.558
  ------------------------------------------------------------------------------------
   mean                   15.21      1.10  0.102                 16.93      6.30  0.566
  (t = clevis origin / pitch pin; rotation is the calibration's rotation error, the same for every frame)

cam_raw_T_base (m), diff-render:
  [ 0.156984 -0.967187  0.199765  0.023991]
  [0.573587 0.253951 0.778787 0.054547]
  [-0.803962 -0.007675  0.594632  0.017398]
  [0. 0. 0. 1.]
================================================================================================================
```

frame_15 (held out: not used by the diff-render fit)

hand-eye (marker only): translation error 15.21 mm rotation error 1.10 deg IoU 0.103

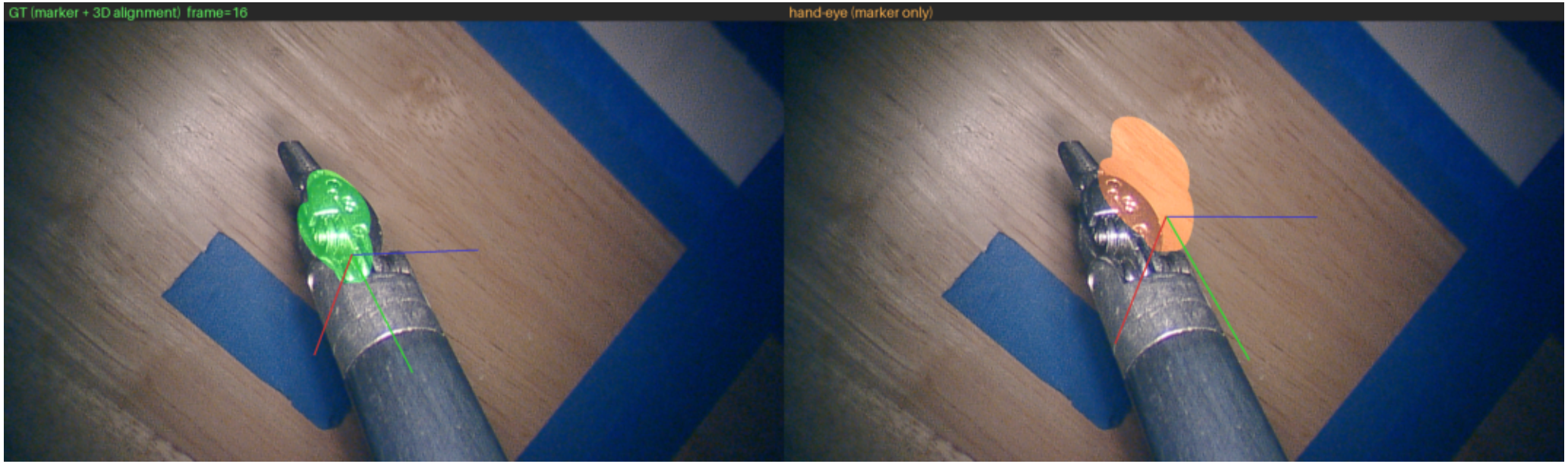


diff-render (markerless): translation error 16.88 mm rotation error 6.30 deg IoU 0.589



frame_16 (held out: not used by the diff-render fit)

hand-eye (marker only): translation error 15.22 mm rotation error 1.10 deg IoU 0.151

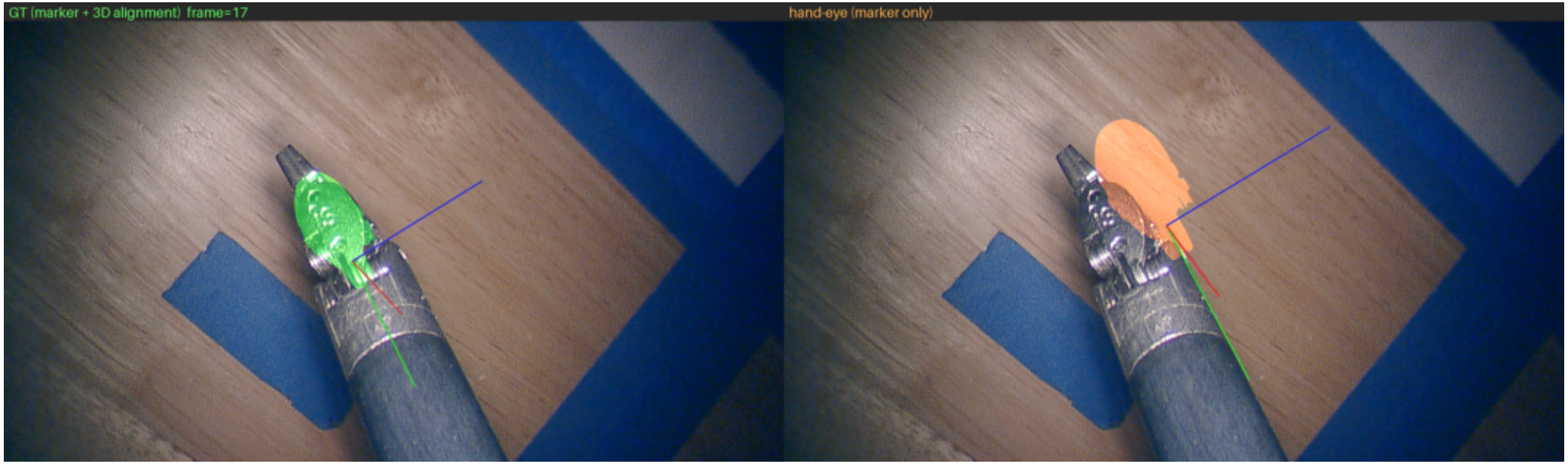


diff-render (markerless): translation error 16.84 mm rotation error 6.30 deg IoU 0.572



frame_17 (held out: not used by the diff-render fit)

hand-eye (marker only): translation error 15.22 mm rotation error 1.10 deg IoU 0.101

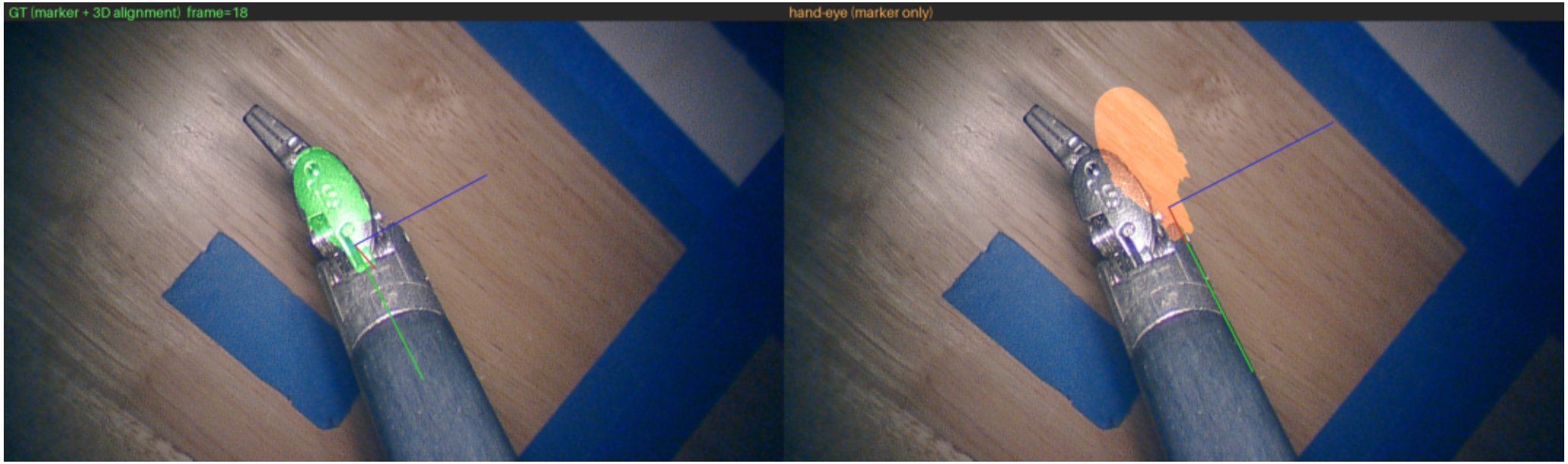


diff-render (markerless): translation error 16.82 mm rotation error 6.30 deg IoU 0.531



frame_18 (held out: not used by the diff-render fit)

hand-eye (marker only): translation error 15.20 mm rotation error 1.10 deg IoU 0.072

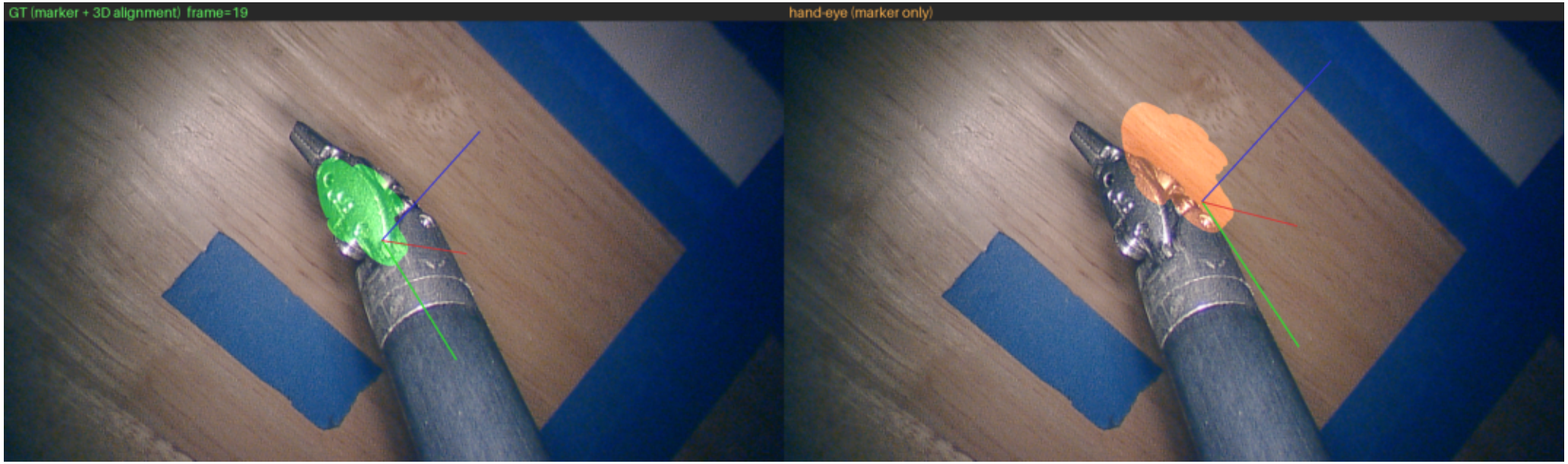


diff-render (markerless): translation error 16.91 mm rotation error 6.30 deg IoU 0.579



frame_19 (held out: not used by the diff-render fit)

hand-eye (marker only): translation error 15.19 mm rotation error 1.10 deg IoU 0.082



diff-render (markerless): translation error 17.20 mm rotation error 6.30 deg IoU 0.558



gt_dataset: mask baseline -- diff-render's generated masks vs hand-drawn clevis segmentation (gt_segmentation_eval.py metrics)

Generated masks: the masks of the diff-render fit in this report (10 training frames with a hand mask), REBUILT -- rebuilt 2026-11-04T16:08:32: the run predates saving its masks; same mask steps, last pass seeded from the run's final calibration instead of its pass-3 estimate (not saved). Hand masks: /home/aseshad4/hand_eye_collection_scripts/gt_dataset/hand_segmentation (clevis only, drawn in gt_segmentation_eval.py's window). IoU = generated vs hand mask over the whole image; 'DR region' = only the pixels the diff-render loss uses (jaw / shaft / ring-prong zones excluded, as in the fit). Pose errors: clevis pose best fitting the generated mask vs best fitting the hand mask (gt_segmentation_eval.py's Nelder-Mead silhouette fit, 4 starts from the diff-render estimate, DR region only), camera frame at the pitch pin; depth = camera z, weakly observable from a silhouette.

frame	IoU	IoU (DR region)	rot err (deg)	trans err (mm)	in-plane (mm)	depth (mm)
0	0.332	0.874	22.665	16.926	2.519	16.737
2	0.166	0.653	30.607	3.277	3.167	0.839
3	0.212	0.841	20.879	28.861	3.210	28.682
5	0.454	0.927	14.295	7.250	1.920	6.991
6	0.276	0.873	19.674	5.921	1.889	5.612
8	0.143	0.674	36.384	42.091	4.351	41.866
9	0.145	0.605	38.670	3.257	2.892	1.499
11	0.450	0.840	22.463	7.357	1.754	7.145
12	0.455	0.894	8.628	7.989	0.170	7.987
14	0.351	0.904	21.983	6.542	1.433	6.384
mean	**0.298**	**0.809**	**23.625**	**12.947**	**2.331**	**12.374**
median	**0.304**	**0.857**	**22.223**	**7.304**	**2.220**	**7.068**
std	**0.122**	**0.112**	**8.820**	**12.184**	**1.099**	**12.503**
max	**0.455**	**0.927**	**38.670**	**42.091**	**4.351**	**41.866**

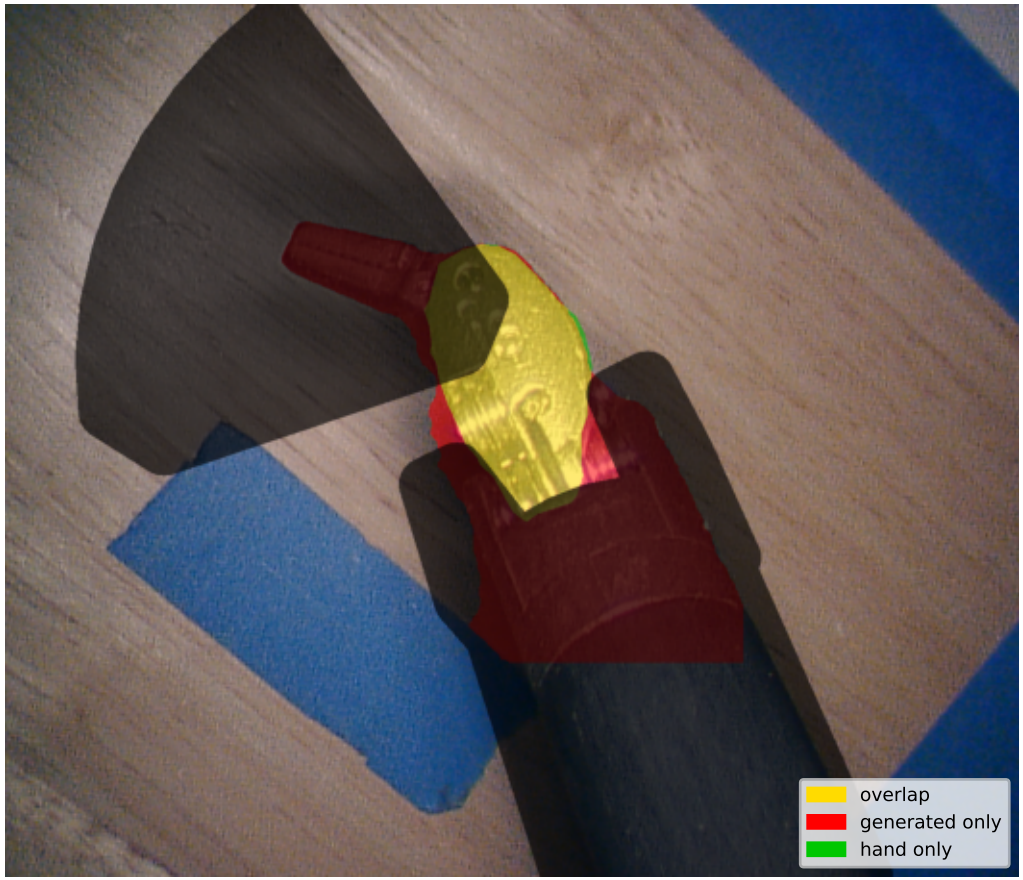
gt_dataset: mask baseline -- what the hand segmentation says about the poses

Context, not a mask metric: the clevis pose best fitting the HAND mask (an image-only reference, as gt_segmentation_eval.py uses it) compared with the ground-truth pose and with the diff-render pose of each frame. Same fit and error definitions as the previous page.

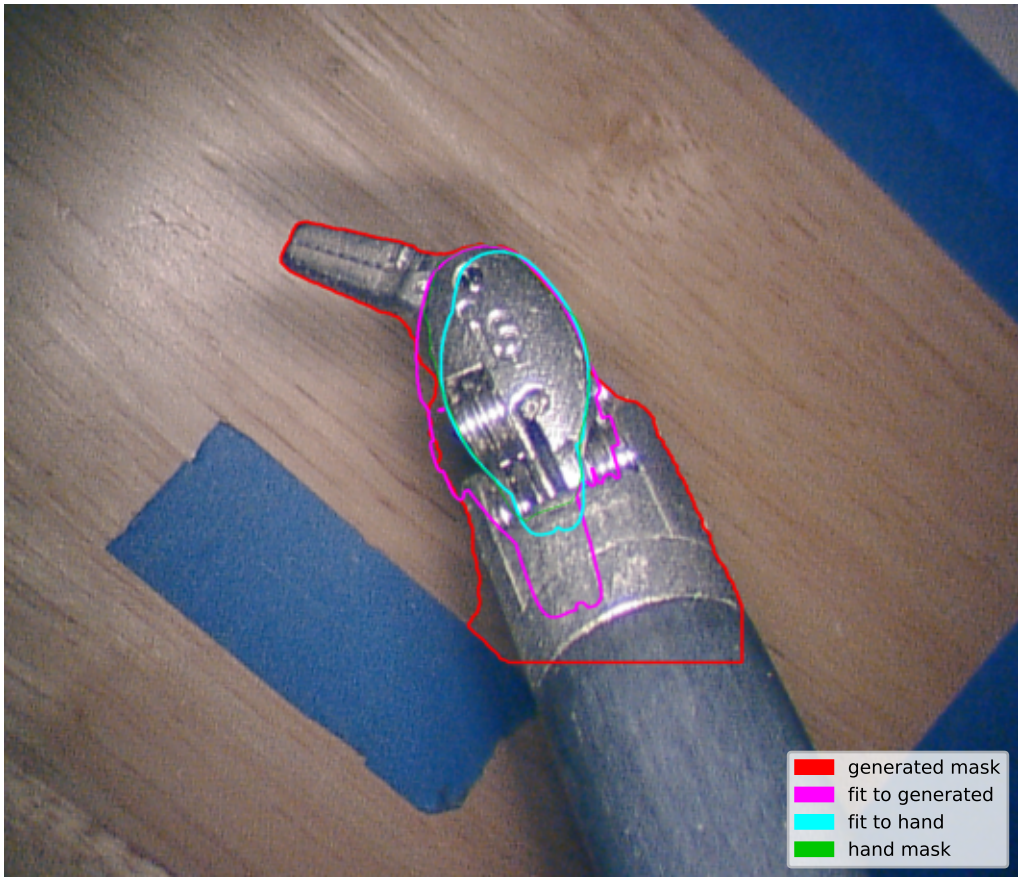
frame	vs GT rot (deg)	vs GT trans (mm)	vs GT in-plane (mm)	vs GT depth (mm)	vs DR rot (deg)	vs DR trans (mm)	vs DR in-plane (mm)	vs DR depth (mm)
0	21.825	3.444	1.333	3.176	27.582	13.830	1.472	13.751
2	32.837	15.925	1.801	15.823	28.544	3.796	3.624	1.133
3	19.844	12.286	4.388	11.476	18.958	28.234	3.173	28.055
5	8.467	11.764	1.852	11.618	13.639	5.512	0.844	5.447
6	11.639	12.788	1.478	12.702	17.726	4.309	1.579	4.009
8	16.779	27.393	3.723	27.139	22.146	44.682	4.042	44.499
9	36.492	17.023	4.494	16.419	30.901	2.043	2.013	0.350
11	28.914	10.623	1.588	10.503	23.865	6.120	1.935	5.806
12	4.277	13.046	2.481	12.808	6.363	3.644	0.356	3.626
14	22.051	9.513	3.679	8.773	26.586	8.230	1.694	8.054
mean	**20.312**	**13.381**	**2.682**	**13.044**	**21.631**	**12.040**	**2.073**	**11.473**
median	**20.835**	**12.537**	**2.166**	**12.160**	**23.006**	**5.816**	**1.815**	**5.627**
std	**9.916**	**5.845**	**1.194**	**5.870**	**7.200**	**13.120**	**1.128**	**13.412**
max	**36.492**	**27.393**	**4.494**	**27.139**	**30.901**	**44.682**	**4.042**	**44.499**

mask baseline, frame 0 (generated mask rebuilt): IoU 0.332 (DR region 0.874)
fit to generated vs fit to hand mask: rot 22.66 deg trans 16.93 mm (x -0.87, y +2.37, z -16.74; in-plane 2.52)

IoU: generated mask over the hand mask (darkened = ignored by the DR loss)

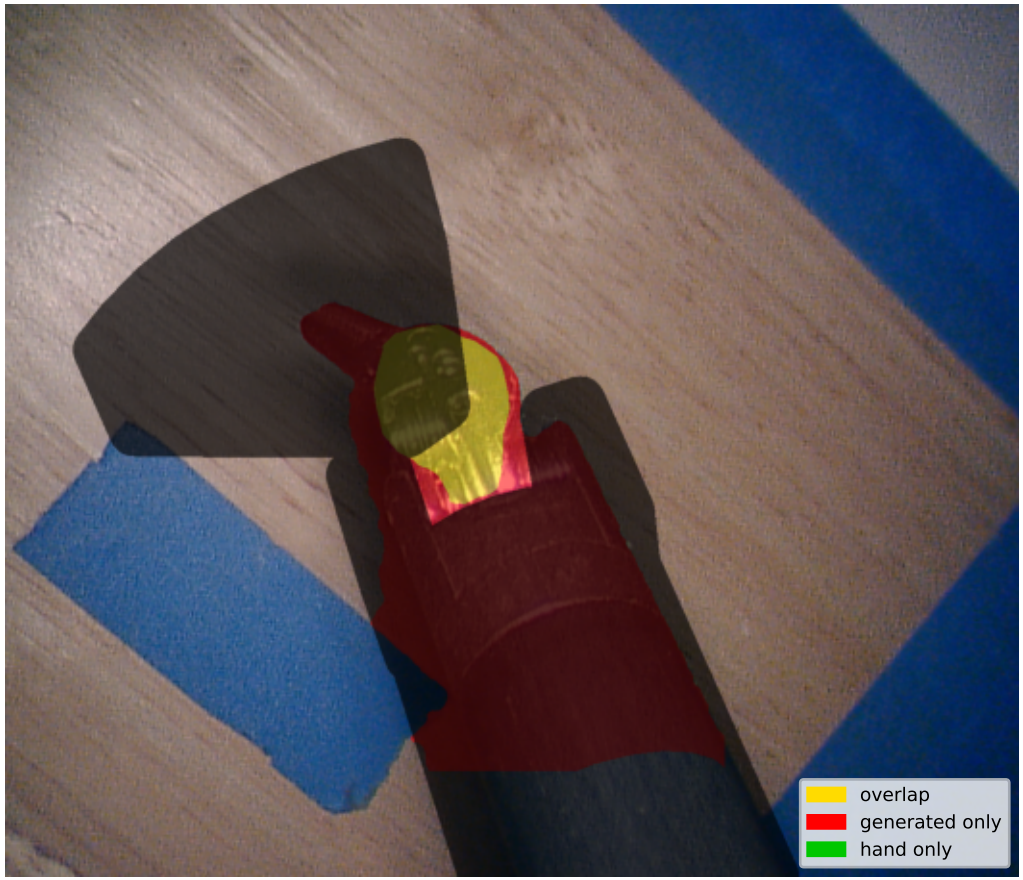


outlines: masks and the clevis poses best fitting each

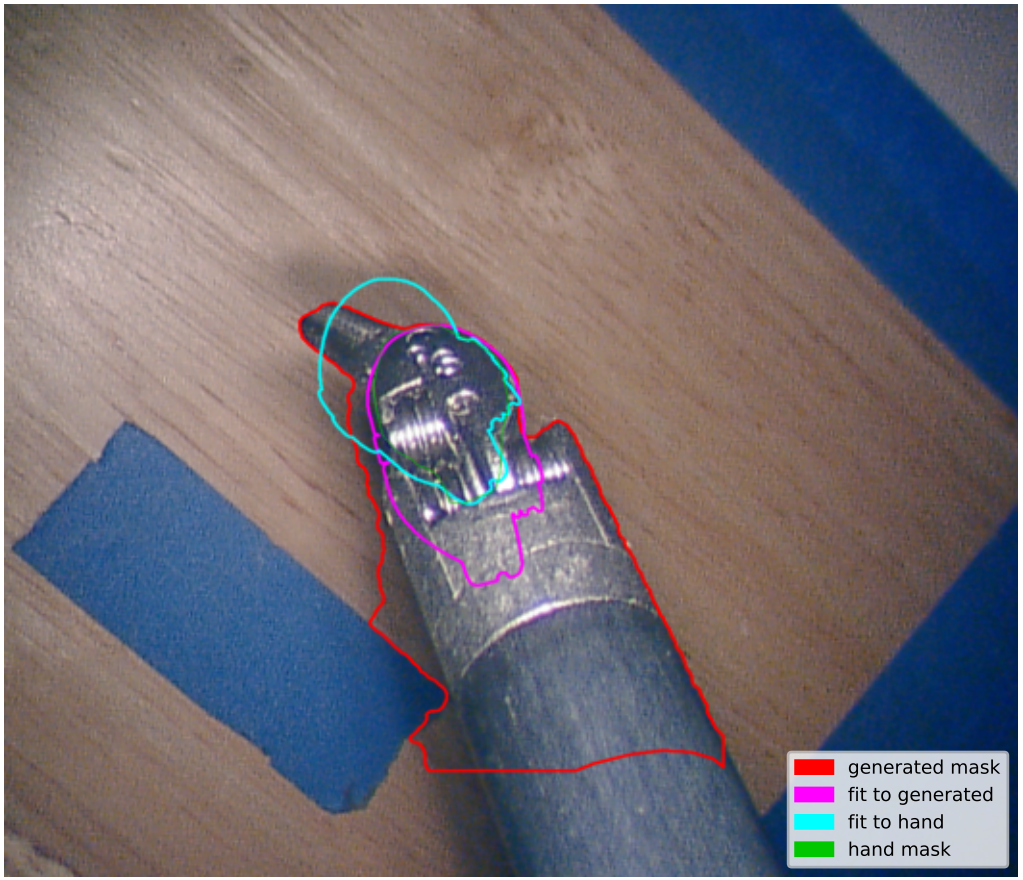


mask baseline, frame 2 (generated mask rebuilt): IoU 0.166 (DR region 0.653)
fit to generated vs fit to hand mask: rot 30.61 deg trans 3.28 mm (x +1.33, y +2.87, z -0.84; in-plane 3.17)

IoU: generated mask over the hand mask (darkened = ignored by the DR loss)

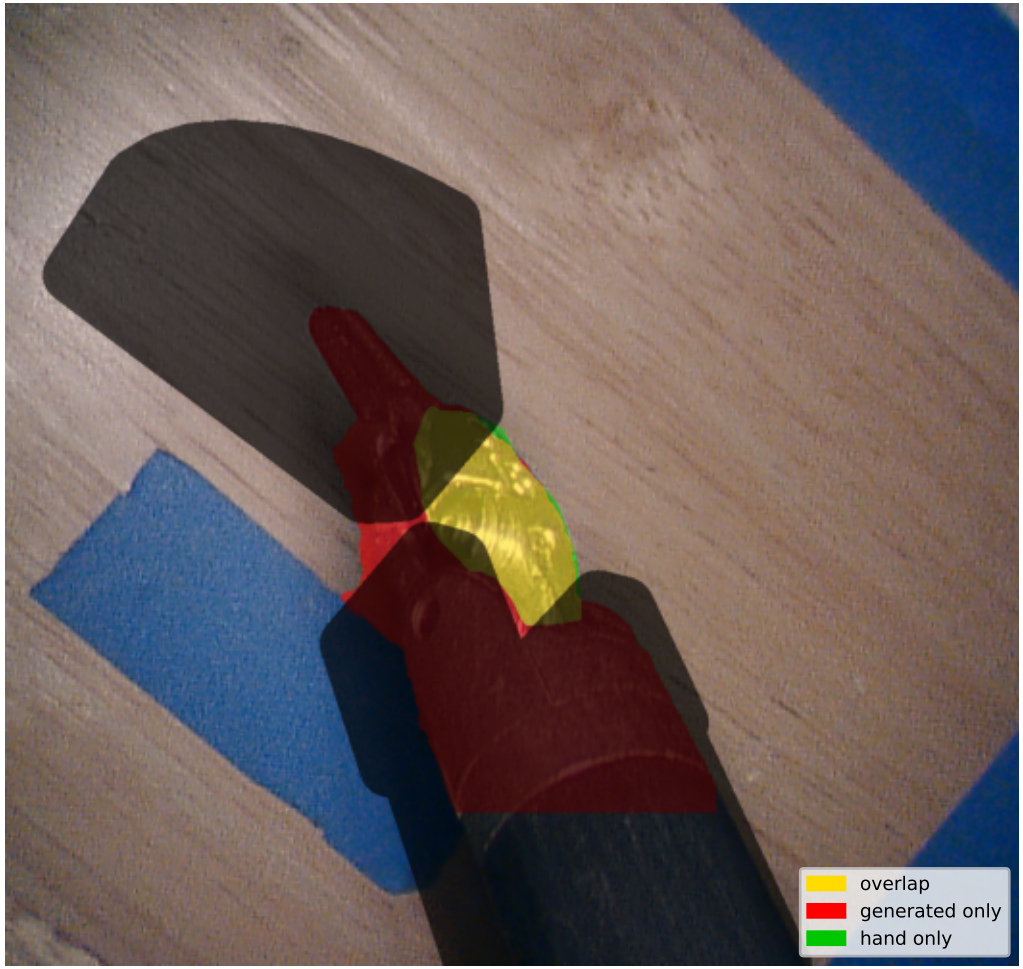


outlines: masks and the clevis poses best fitting each

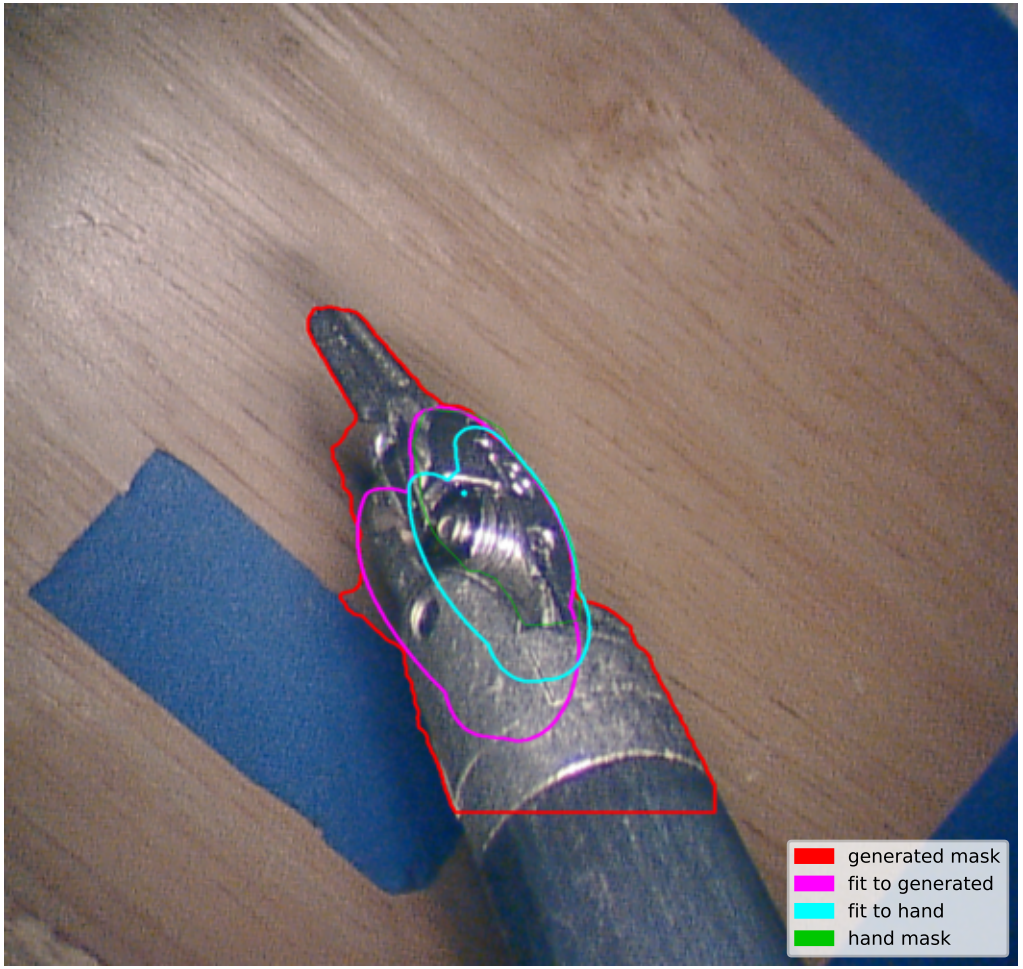


mask baseline, frame 3 (generated mask rebuilt): IoU 0.212 (DR region 0.841)
fit to generated vs fit to hand mask: rot 20.88 deg trans 28.86 mm (x -3.15, y -0.60, z -28.68; in-plane 3.21)

IoU: generated mask over the hand mask (darkened = ignored by the DR loss)

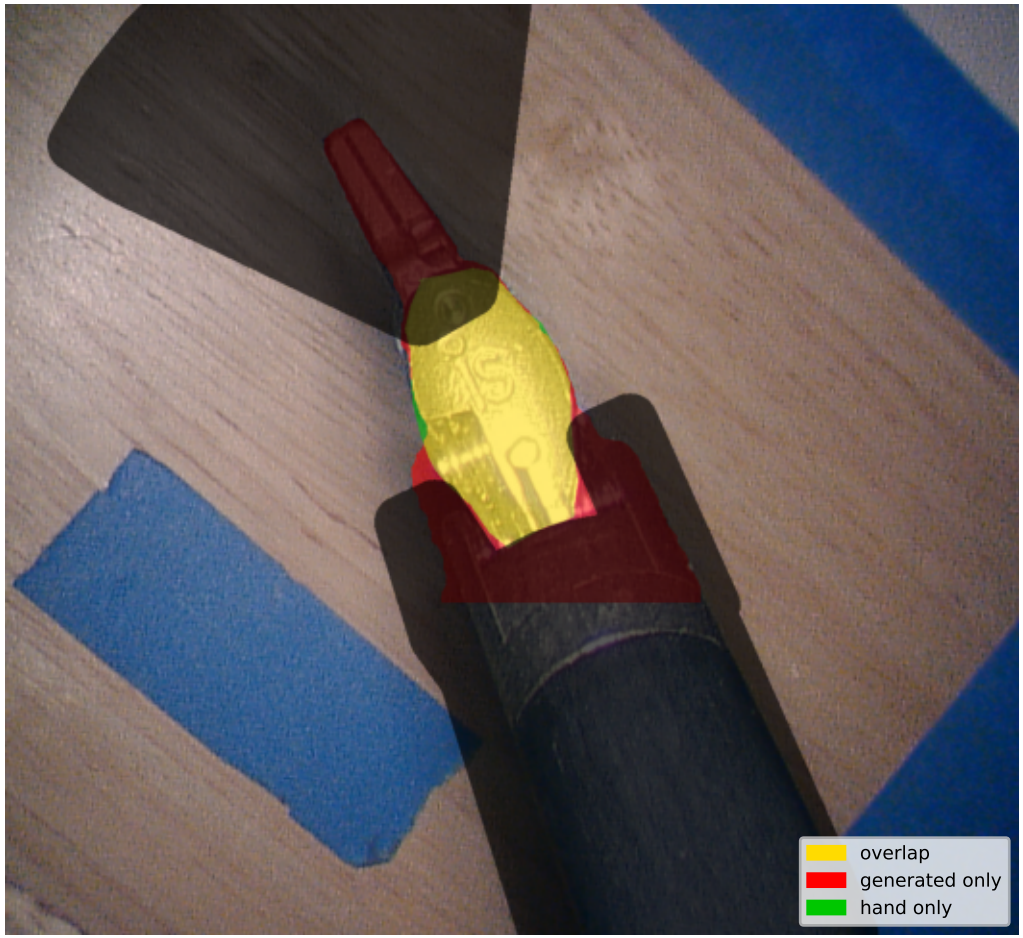


outlines: masks and the clevis poses best fitting each

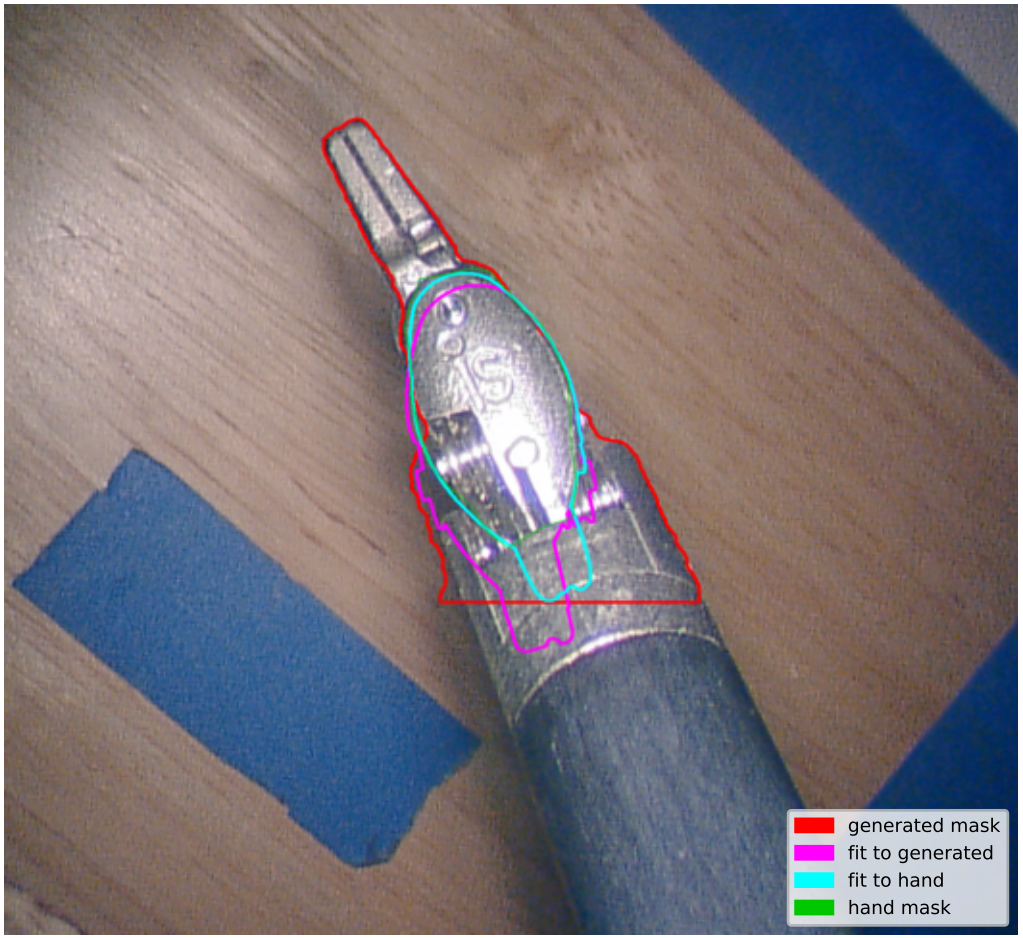


mask baseline, frame 5 (generated mask rebuilt): IoU 0.454 (DR region 0.927)
fit to generated vs fit to hand mask: rot 14.29 deg trans 7.25 mm (x -1.26, y +1.45, z -6.99; in-plane 1.92)

IoU: generated mask over the hand mask (darkened = ignored by the DR loss)

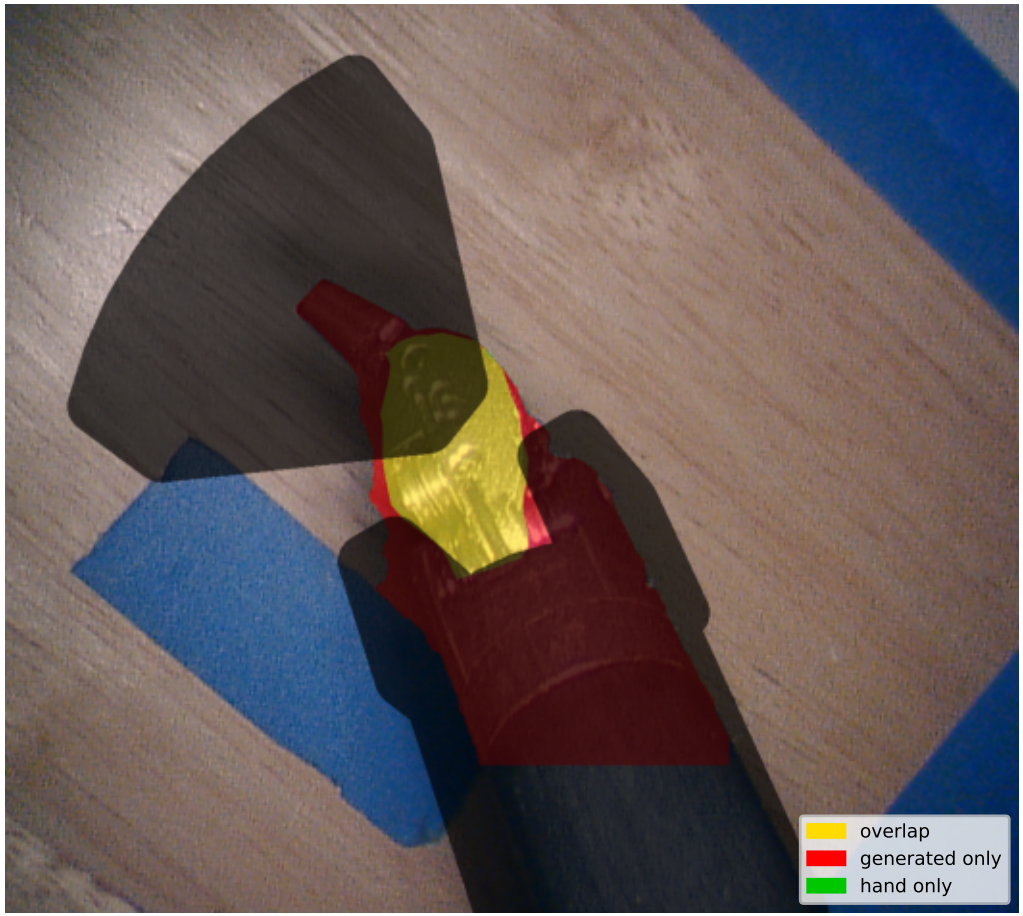


outlines: masks and the clevis poses best fitting each

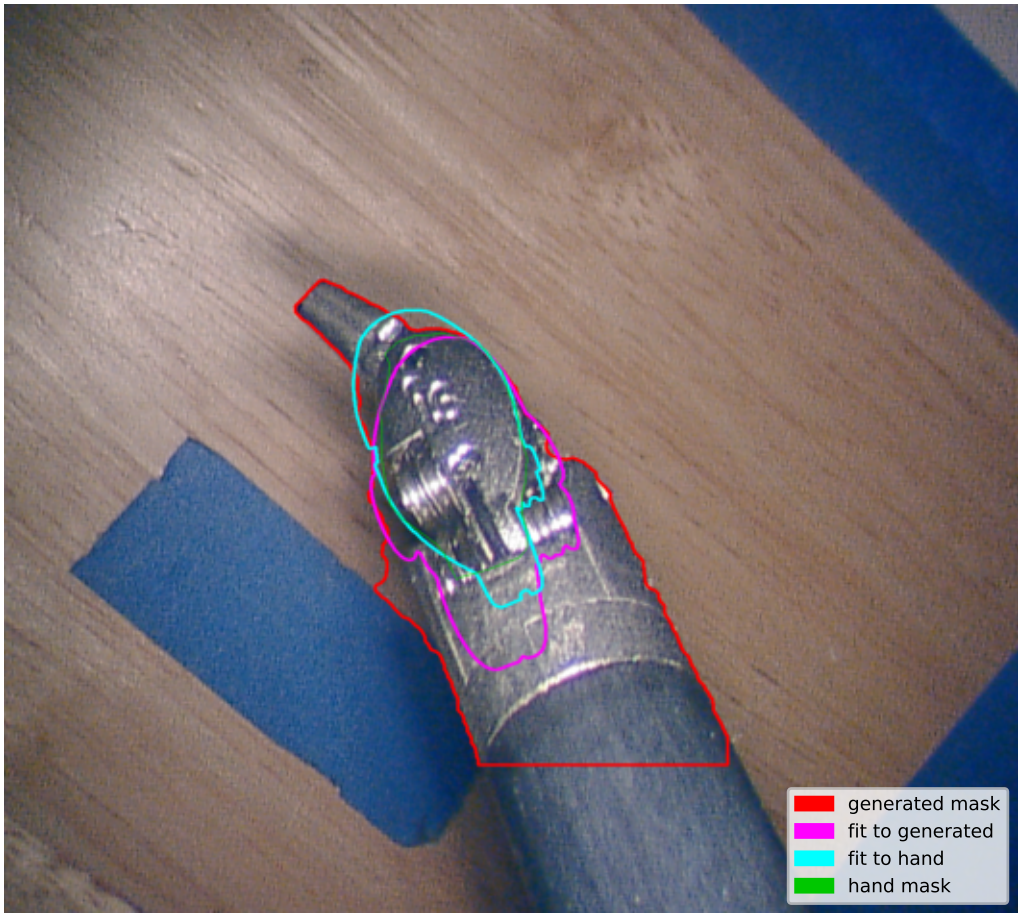


mask baseline, frame 6 (generated mask rebuilt): IoU 0.276 (DR region 0.873)
fit to generated vs fit to hand mask: rot 19.67 deg trans 5.92 mm (x -0.35, y +1.86, z -5.61; in-plane 1.89)

IoU: generated mask over the hand mask (darkened = ignored by the DR loss)

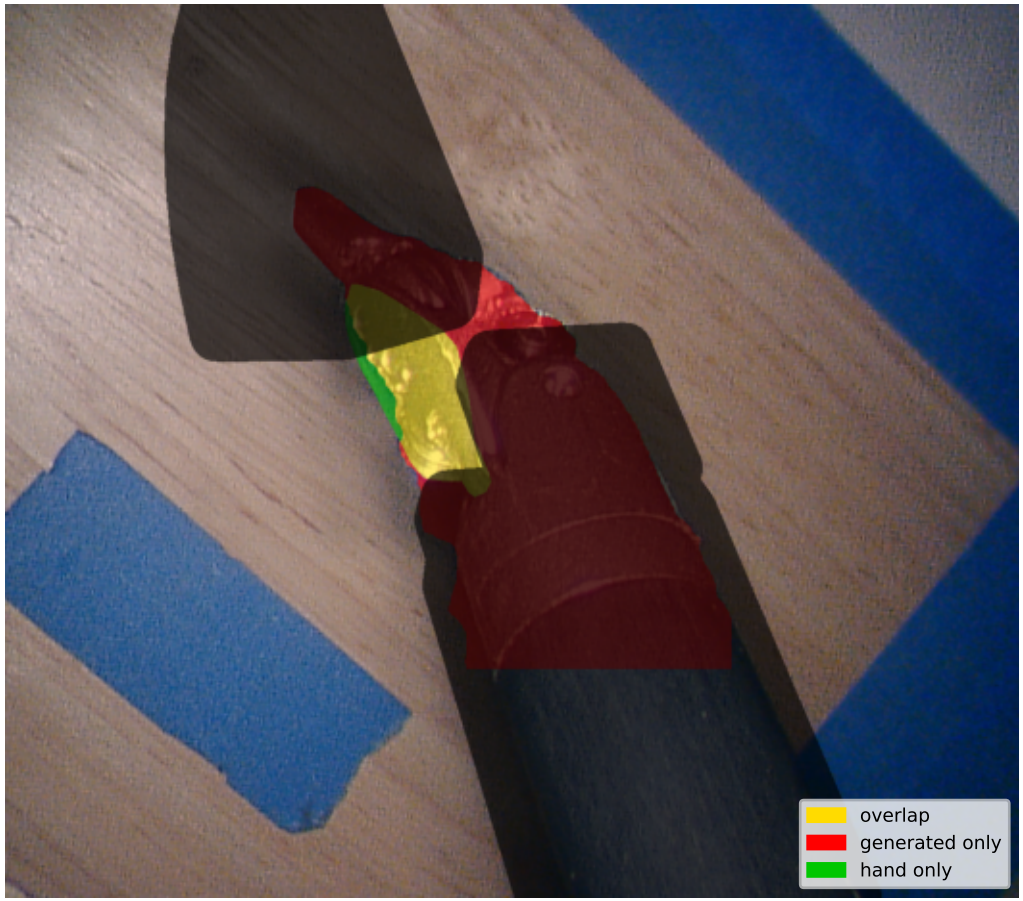


outlines: masks and the clevis poses best fitting each

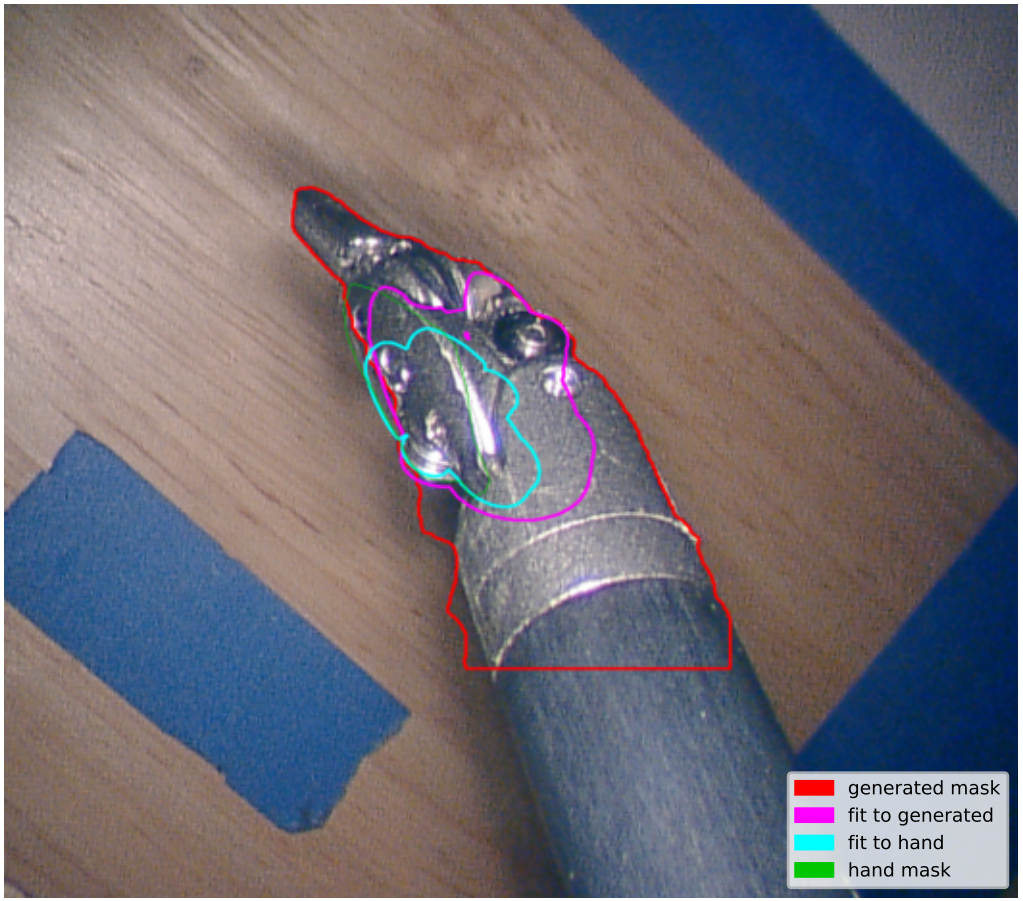


mask baseline, frame 8 (generated mask rebuilt): IoU 0.143 (DR region 0.674)
fit to generated vs fit to hand mask: rot 36.38 deg trans 42.09 mm (x -4.33, y +0.42, z -41.87; in-plane 4.35)

IoU: generated mask over the hand mask (darkened = ignored by the DR loss)

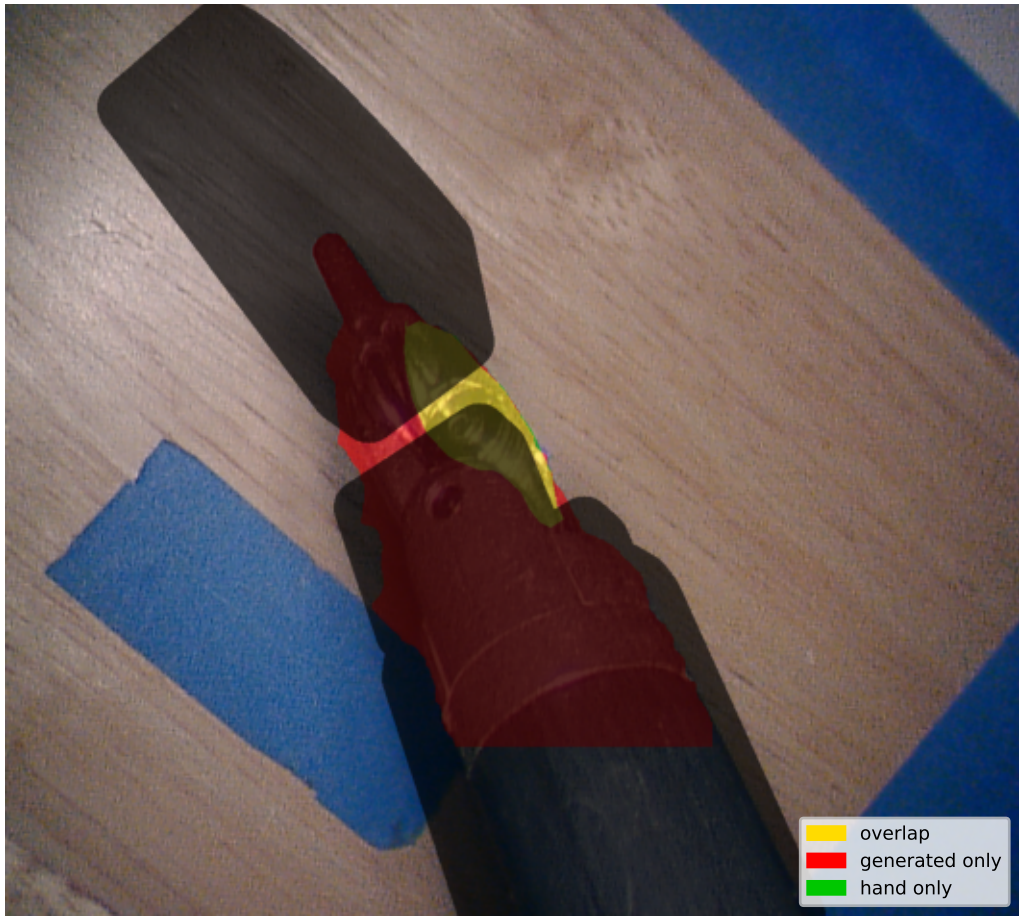


outlines: masks and the clevis poses best fitting each

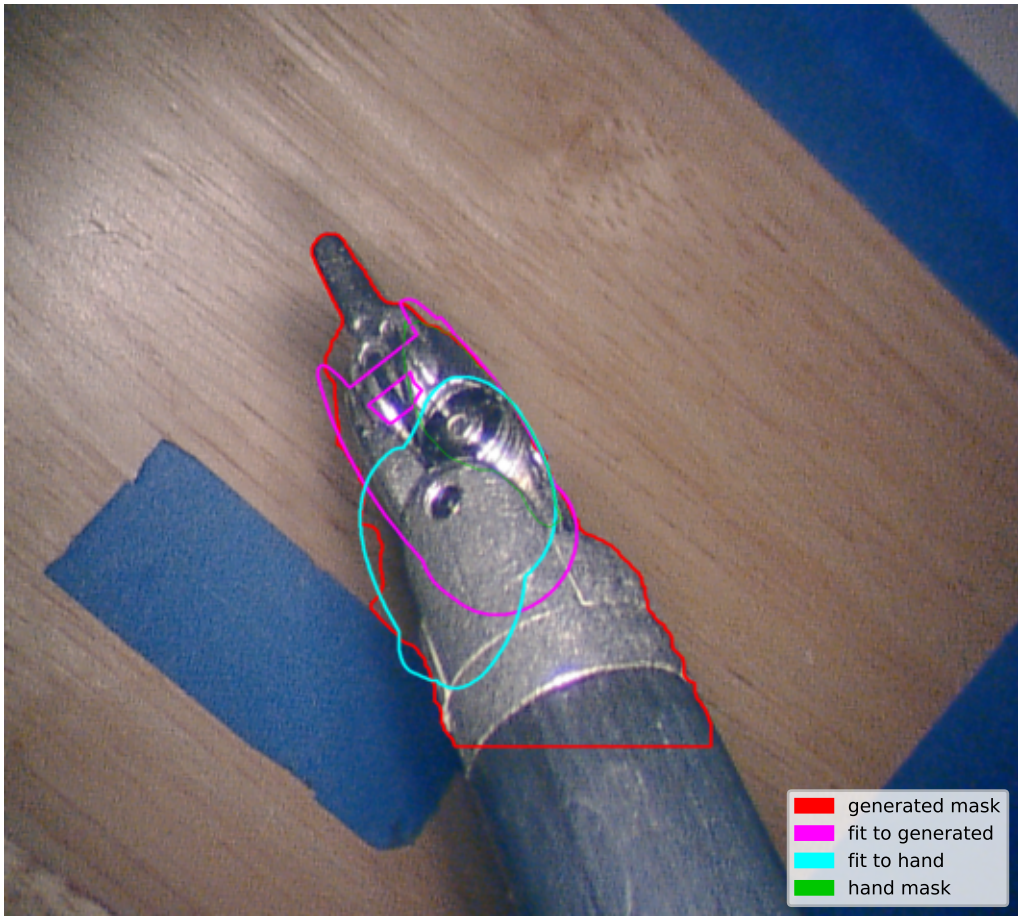


mask baseline, frame 9 (generated mask rebuilt): IoU 0.145 (DR region 0.605)
fit to generated vs fit to hand mask: rot 38.67 deg trans 3.26 mm (x +1.43, y -2.51, z +1.50; in-plane 2.89)

IoU: generated mask over the hand mask (darkened = ignored by the DR loss)

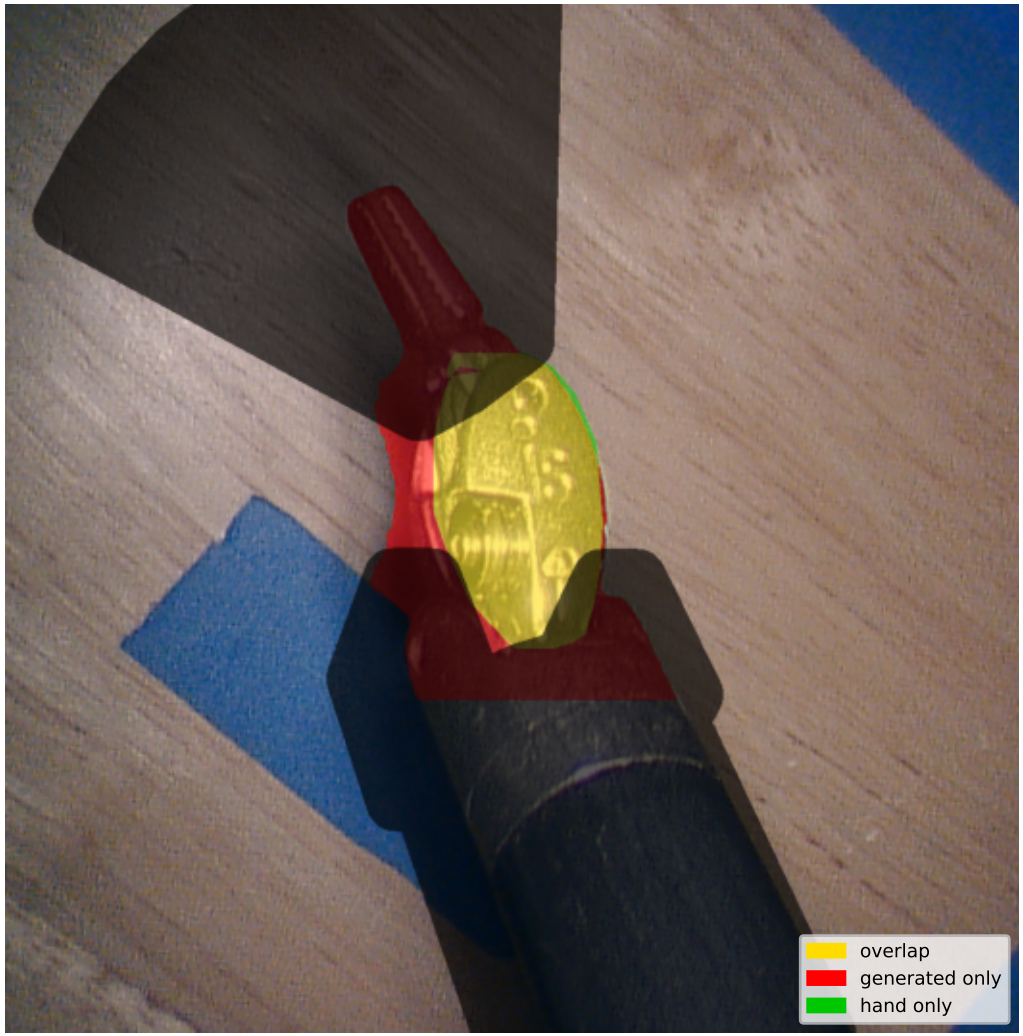


outlines: masks and the clevis poses best fitting each

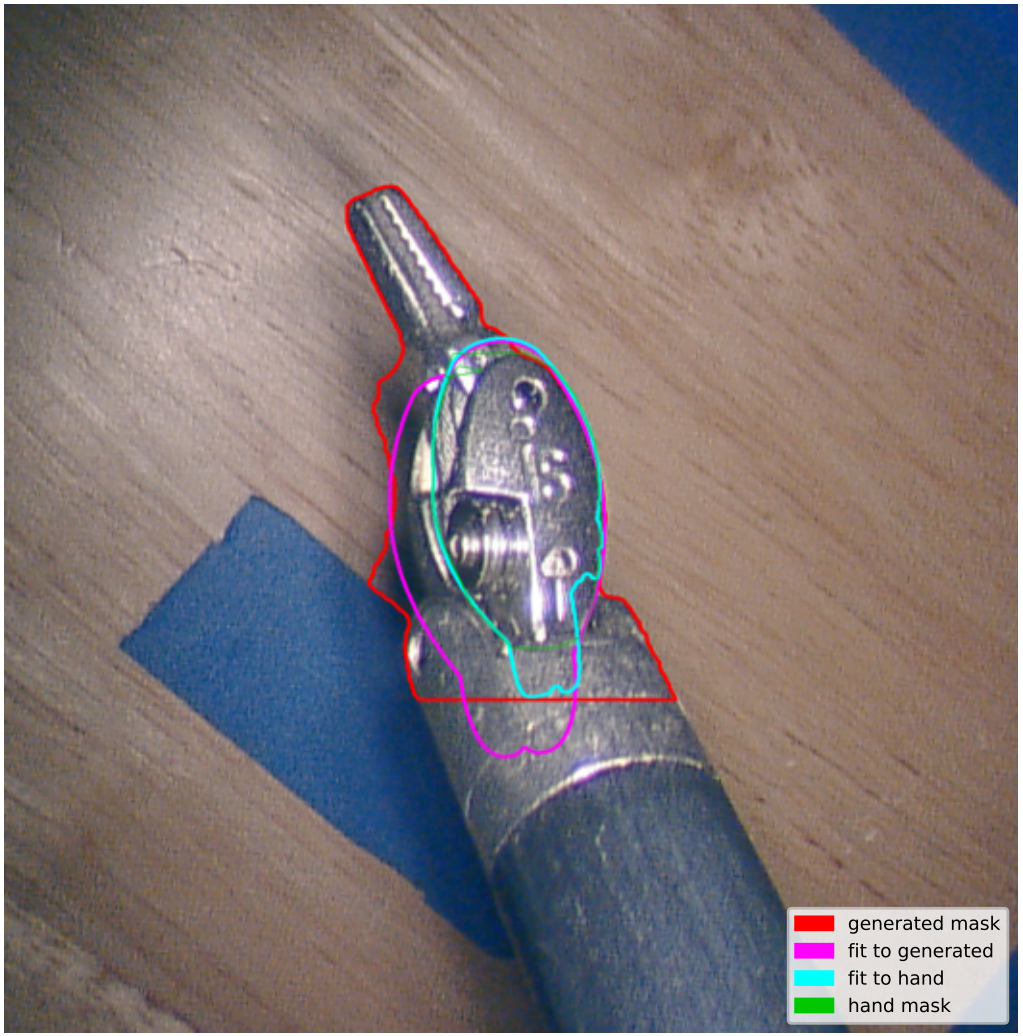


mask baseline, frame 11 (generated mask rebuilt): IoU 0.450 (DR region 0.840)
fit to generated vs fit to hand mask: rot 22.46 deg trans 7.36 mm (x -0.91, y +1.50, z -7.15; in-plane 1.75)

IoU: generated mask over the hand mask (darkened = ignored by the DR loss)

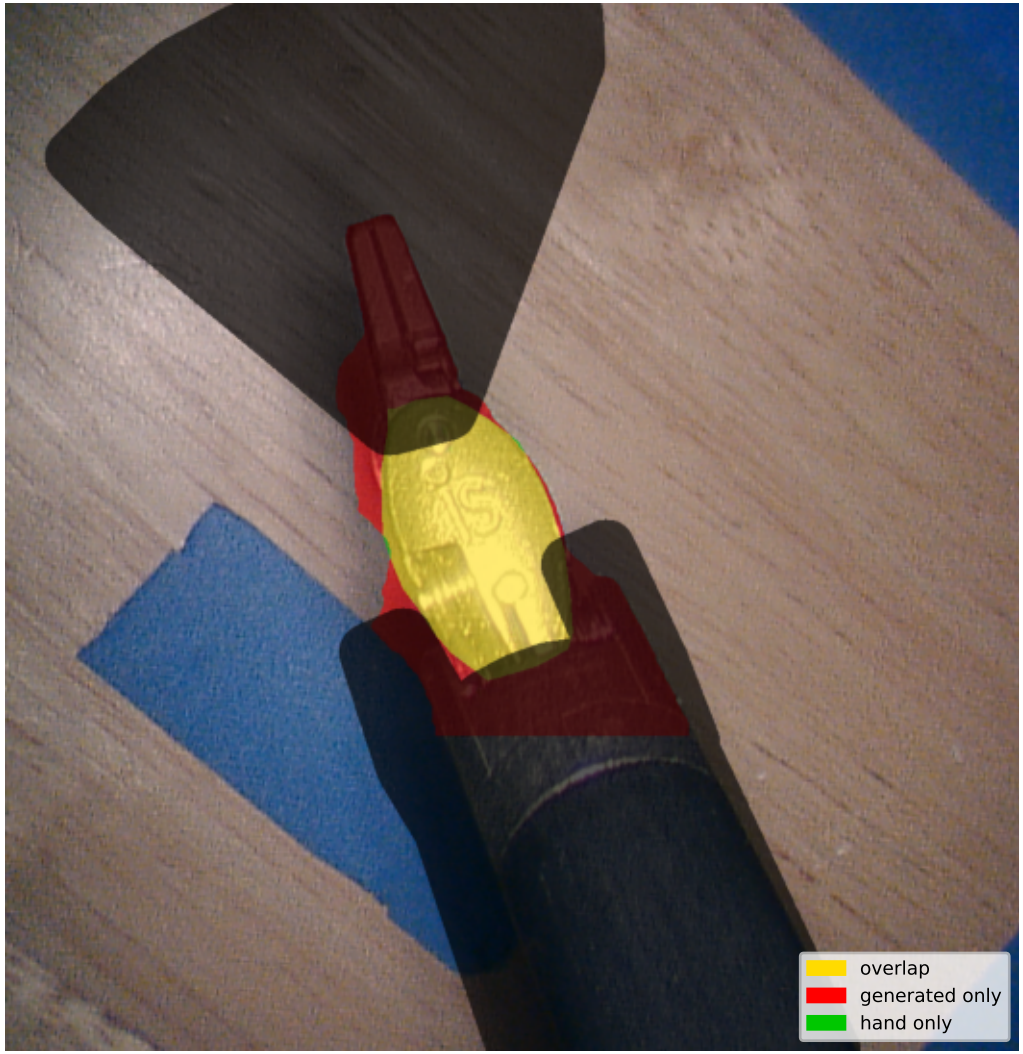


outlines: masks and the clevis poses best fitting each



mask baseline, frame 12 (generated mask rebuilt): IoU 0.455 (DR region 0.894)
fit to generated vs fit to hand mask: rot 8.63 deg trans 7.99 mm (x +0.06, y -0.16, z -7.99; in-plane 0.17)

IoU: generated mask over the hand mask (darkened = ignored by the DR loss)

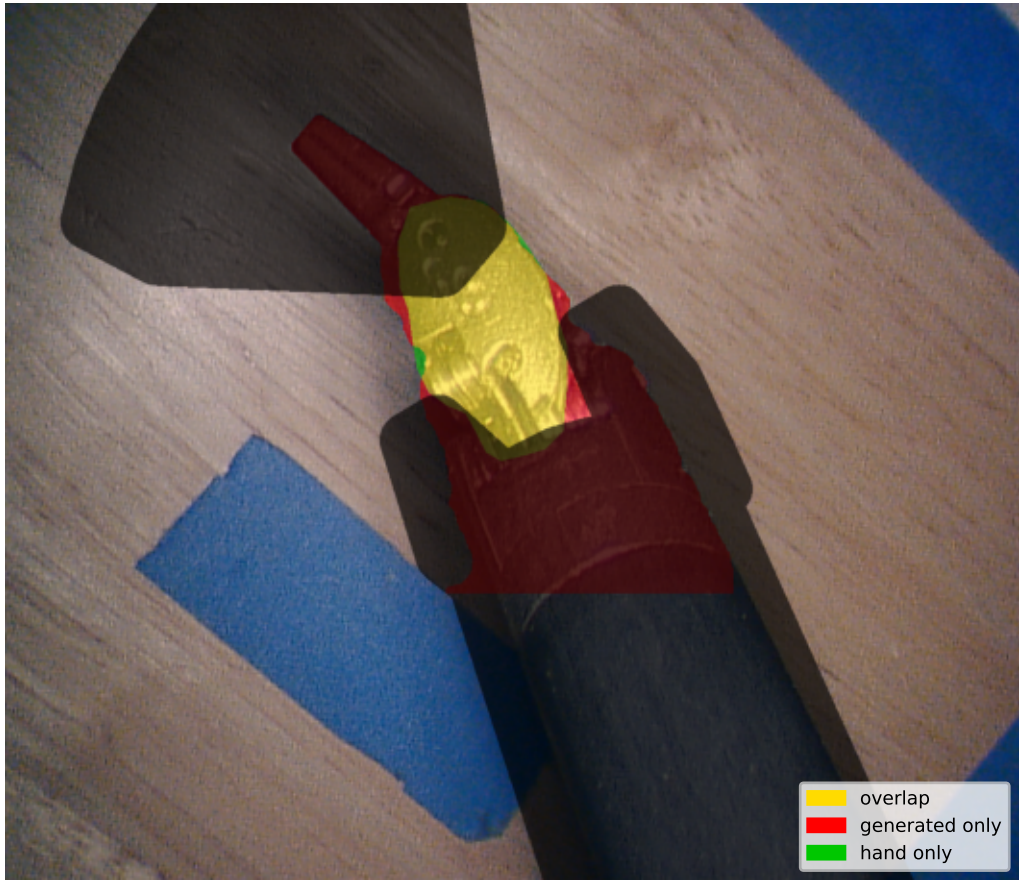


outlines: masks and the clevis poses best fitting each



mask baseline, frame 14 (generated mask rebuilt): IoU 0.351 (DR region 0.904)
fit to generated vs fit to hand mask: rot 21.98 deg trans 6.54 mm (x +1.41, y +0.24, z -6.38; in-plane 1.43)

IoU: generated mask over the hand mask (darkened = ignored by the DR loss)



outlines: masks and the clevis poses best fitting each

